

Vansh Gupta

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Professional Summary

Computer Science graduate building production-grade GenAI systems — multi-agent RAG pipelines, LLM-based reasoning agents, and document-intelligence tools using LangChain, LangGraph, and LangSmith.

Education

Birla Institute of Technology and Science, Pilani
Dubai, UAE 2021-2025

G.D. Goenka Public School
Score: 94.7% Indore, India

Skills

Programming: Python, JavaScript, SQL

Web Development: React.js, Node.js, REST APIs, HTML5, CSS, MySQL, PostgreSQL

Machine Learning: PyTorch, LangChain, LangGraph, LangSmith, Scikit-learn, Deep Learning, LLMs, Pandas

Projects

V-RAG — Verified Retrieval-Augmented Generation

August 2026

Python, LangChain, LangGraph, LangSmith, OpenAI, Chroma, Pydantic

- Developed a multi-agent RAG system for evidence grounded question answering over a local legal document corpus, which included Responder, Prosecutor, Investigation, and Judge agents to detect unsupported claims and iteratively revise responses.
- Improved legal retrieval by enriching chunk embeddings with case-level metadata, achieving a 44-position improvement in target-chunk retrieval rank (45 → 1) in evaluation; built a Streamlit interface with execution tracing and LangSmith observability.

Compliance Agent — Intelligent Document Compliance & Automation Agent

May 2025

Python, LangChain, Gemini API, PostgreSQL, OCR, PDF Parsing

- Designed an AI-powered agent capable of reasoning across structured and unstructured documents to ensure regulatory compliance.
- Implemented robust document ingestion pipeline using OCR (Google Vision) and PDF parsing to extract structured data.
- Developed multi-step reasoning workflows using LangChain and Gemini to automate compliance checks and generate insights.
- Built modular components for rule-based filtering, entity extraction, and multi-document correlation.

Nasa Space App Challenge — Space Tourism Educational Game

December 2024

HTML, CSS, JavaScript, Python

- Designed and developed a web-based educational game simulating space tourism experiences, aimed at raising public interest and awareness about space exploration.
- Engineered engaging user interfaces and interactive elements using HTML, CSS, and JavaScript for a seamless gamified experience.
- Collaborated in a multidisciplinary team to refine user experience and incorporate educational content aligned with NASA's outreach goals.
- Awarded 1st Rank in UAE and selected as a Global Nominee, demonstrating innovation, technical excellence, and creative problem-solving at an international level.

HOPE— AI Powered Mental Health Platform

2022-2023

HTML, CSS, JavaScript, OpenAI, Whisper API

- Built a full-stack web platform providing CBT (Cognitive Behavioural Therapy) therapy sessions using AI-driven chatbots.
- Integrated OpenAI's GPT and Whisper APIs for natural language conversations and voice input/output, increasing accessibility for users, developed additional features including a habit tracker, guided meditation modules, and relaxation music playlists to support mental health.
- Achieved 60+ monthly active users; project was backed by Microsoft for Startups after submission for GDSC Solutions Challenge.

Experience

Freelance

September 2025 – Present

- Built and customized MMABOXING a Shopify e-commerce storefront for a combat-sports brand, translating business requirements into a production-ready online store.
- Developed Flasseq, a full-stack clothing e-commerce platform using EverShop as a React theme and Razorpay.

Aseries Envirotek India Pvt. Ltd.

Remote

Software Developer Intern

February 2025 – September 2025

- Developed a web-based portal using React.js, Node.js, and PostgreSQL to automate the generation of Environmental Impact Assessment (EIA) reports.
- Engineered dynamic document generation by integrating placeholder replacement in official forms (Form 1, Form 1A, Conceptual Plan, EMP) while preserving original formatting for PDF and DOC outputs.
- Utilizing cPanel for deployment management and Doc processing libraries for document automation.
- Reduced manual documentation efforts and improving project tracking for client submissions by building an intuitive admin dashboard.
- Worked independently to drive project architecture, backend API development, and frontend integration, ensuring scalability and maintainability.

Trajectorie

Remote

Frontend Developer Intern

September 2023 – July 2024

- Led frontend development for a quizzing platform using React.js, delivering an improvement in responsiveness and user experience.
- Integrated RESTful APIs for seamless frontend-backend communication, enhancing system reliability and cross-browser compatibility.

Digiproctor

Remote

Software Developer Intern

June 2023 – August 2023

- Built an internal interviewing platform using JavaScript and Node.js, streamlining onboarding and improving efficiency.
- Developed and integrated a Rasa chatbot, reducing customer query resolution time and enhancing support satisfaction.
- Engineered a dynamic browser-based PDF editor, boosting user engagement.

Leadership

Google Developer Student Club

Dubai

Technical Executive

August 2023 – June 2024

- Led the Web Development network under GDSC, mentoring students and hosting technical workshops.
- Conducted Introduction to HTML, CSS, and JavaScript workshops attended by 60+ students.
- Organized multiple Capture the Flag (CTF) competitions, engaging 300+ university participants.

Awards

- NASA International Space Apps Challenge (2023) — UAE Winner and Global Nominee.
- Emirati Sign Language Project — 2nd Place, Winner of 2000 AED Prize.
- ACM-CIIED University Ideathon — Winner, awarded 500 AED.
- RIFT CTF (Google Developer Groups 2025) — 3rd Place.
- Exploit3rs CTF (UAE) — 5th Rank out of 120 teams.
- BITS Pilani Postman API Hackathon — 9th Rank out of 2300 teams.